

Best-in-Class Cyber Security Operations Platform

Master Product, Functional and Non-Functional Requirements Document

Document purpose

This document is part of a build-and-operate pack for a modular Cyber Security Operations Platform capable of SOC-as-a-Service, MDR, Managed XDR, sovereign SOC, enterprise private SOC, and MSSP white-label deployments.

1. Executive product vision

The platform is a modular Cyber Security Operations Platform, not a single-country SOC dashboard. It is intended to power SOC-as-a-Service, MDR, Managed XDR, sovereign/local SOC deployments, enterprise private SOC, and MSSP white-label offerings. The platform will provide native SOC functions while integrating with customer security controls and IT systems.

Strategic design principle

Build a native SOC operating layer first, then add external integrations in controlled phases. The product must work with basic syslog/API/webhook feeds and become more powerful as Microsoft, EDR, cloud, firewall, identity, ticketing and collaboration systems are connected.

2. Target customer segments and use cases

Segment	Primary need	Platform proposition
SMEs and mid-market	Affordable 24/7 monitoring without building a SOC	Essential Monitor and Standard SOCaaS with Microsoft, firewall syslog, endpoint and reporting.
Banks and fintechs	Identity, endpoint, cloud and payment-adjacent monitoring with compliance evidence	Advanced MDR/Managed XDR with strict audit trails, SLA, incident evidence and executive reporting.
Telecoms and utilities	High-volume monitoring, critical infrastructure resilience and incident coordination	Critical Infrastructure tier with custom detections, senior escalation, IR retainer and sector dashboard.
Government and regulators	Sovereign data residency, compliance evidence and sector oversight	Sovereign SOC deployment with country-specific compliance packs and optional central buyer dashboard.
Enterprise private SOC	Dedicated SOC platform for in-house security team	Private tenant/instance with admin, analyst workbench, playbooks and integrations.
MSPs/MSSPs	White-label security operations platform	Partner portal, customer sub-tenancy, branding, reseller billing and shared analyst workflows.

3. Deployment models

Model	Description	When to use
Shared SaaS	Multi-tenant cloud platform operated by the provider.	SMEs, mid-market, rapid onboarding, lower cost.
Sovereign cloud / local hosting	Data-resident platform deployed in-country or approved jurisdiction.	Government, CII, regulated sectors, public-sector anchor clients.
Dedicated customer instance	Single-tenant instance with isolated storage and optional dedicated compute.	Banks, telecoms, large enterprises, data sovereignty requirements.
Hybrid collector model	Customer-side collector forwards normalized events/alerts to the platform.	Clients with sensitive data, bandwidth constraints, or internal log retention requirements.
White-label MSSP mode	MSSP can onboard and manage its own customers under its brand.	Channel expansion and regional partner scaling.

4. Product modules

Module	Core requirements
Tenant and customer management	Customer records, service tier, retention, data residency, SLAs, contacts, escalation matrix, billing terms and integration inventory.
Identity, access and permissions	RBAC/ABAC, MFA, SSO, privileged admin roles, analyst roles, customer roles, API tokens and separation of duties.
Alert ingestion and normalization	APIs, syslog, webhooks, file ingestion, connector polling, schema mapping, deduplication and enrichment queues.
Analyst workbench	Alert queue, severity, customer context, timeline, entity graph,

	evidence, notes, task assignment and escalation.
Incident case management	Case creation, classification, impacted entities, severity, SLA, status, evidence, approvals, customer communication and closure report.
Detection engineering	Rule catalogue, rule lifecycle, Sigma support, MITRE mapping, tuning, false-positive feedback and coverage heatmaps.
SOAR and playbooks	Approval-based response workflows, containment actions, notification templates, ticket sync and post-action audit.
Threat intelligence	IOC enrichment, TLP handling, STIX/TAXII support, MISP/OTX/commercial feed integration and watchlists.
AI SOC assistant	Alert summarisation, correlation recommendations, attack timeline, report drafting, detection tuning suggestions and analyst copiloting.
Reporting and compliance evidence	Executive dashboards, monthly reports, SLA reports, evidence packs, regulator-ready incident summaries and audit exports.
MSP/white-label portal	Partner hierarchy, delegated administration, branded reports, reseller pricing and sub-customer management.
Billing and subscription	Service tiers, log volume entitlements, overage tracking, add-ons, invoices and revenue reporting.

5. Functional requirements

ID	Area	Requirement	Release
FR-001	Customer onboarding	The platform shall provide configurable customer onboarding functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-002	Asset inventory	The platform shall provide configurable asset inventory functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-003	Log ingestion	The platform shall provide configurable log ingestion functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-004	Alert management	The platform shall provide configurable alert management functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-005	Incident management	The platform shall provide configurable incident management functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-006	Investigation timeline	The platform shall provide configurable investigation timeline functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-007	Evidence handling	The platform shall provide configurable evidence handling functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	MVP
FR-008	SLA management	The platform shall provide configurable sla management functionality with tenant-aware permissions, audit trail, lifecycle	MVP

		status, and export/API support.	
FR-009	SOAR playbooks	The platform shall provide configurable soar playbooks functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	V1
FR-010	Reporting	The platform shall provide configurable reporting functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	V1
FR-011	Compliance packs	The platform shall provide configurable compliance packs functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	V1
FR-012	Admin configuration	The platform shall provide configurable admin configuration functionality with tenant-aware permissions, audit trail, lifecycle status, and export/API support.	V1

6. Non-functional requirements

ID	Category	Requirement
NFR-001	Security	All tenant data must be isolated logically and protected with encryption in transit and at rest.
NFR-002	Availability	Production service target should be 99.9%+ for standard tiers and higher for critical infrastructure/dedicated deployments.
NFR-003	Auditability	All administrative, analyst, AI, integration and playbook actions must create immutable audit logs.
NFR-004	Scalability	Architecture must scale by tenant, event volume, connector volume, analyst users and report generation.
NFR-005	Performance	Priority alerts should appear in the analyst queue within agreed latency after receipt/collection.
NFR-006	Data residency	Tenant-level residency and retention policies must be configurable by deployment and contract.
NFR-007	Observability	Platform components must emit metrics, logs and traces to monitor own platform security and reliability.
NFR-008	Extensibility	Connector framework must allow new source/action integrations without rebuilding the core platform.
NFR-009	Safety	Automated containment must be controlled by tenant-specific authority-to-act and approval rules.
NFR-010	Maintainability	All major modules must include automated tests, API documentation and deployment scripts.

7. Service tiers and entitlement design

Tier	Typical customer	Included scope	Hard limits to define contractually
Essential Monitor	Small agency/SME	Core alert receipt, Microsoft 365, basic endpoint/firewall syslog,	Assets, log volume, retention, included integrations, report

		monthly report.	frequency.
Standard SOCaaS	Mid-market/MDA	24/7 monitoring, identity, endpoint, server/firewall logs, case management, quarterly review.	SLA window, retention, included playbooks, volume caps.
Advanced MDR	Regulated entity	EDR/XDR integrations, threat intel, response playbooks, tabletop exercise, compliance pack.	IR hours, detection customization, number of integrations.
Critical Infrastructure	Bank/telco/utility	Priority queue, custom detection, senior analyst escalation, board dashboard, IR retainer.	Authority-to-act, escalation times, data residency, retention.
Enterprise/Sovereign/White-label	Cluster buyer/MSSP/large enterprise	Dedicated/sovereign/white-label options, central dashboard, multi-year commitments.	Custom pricing, onboarding factory, scope boundaries, support model.

8. MVP definition

The MVP should prove the platform can onboard tenants, ingest alerts/logs, triage security events, create incident cases, provide analyst and customer visibility, and generate credible reports. The MVP should focus on a narrow integration set rather than attempting every connector.

MVP capability	Minimum acceptance outcome
Tenant portal and admin portal	Provider can create customers, users, service tier, SLAs and escalation contacts.
Ingestion API/webhook/syslog	Platform can receive test alerts/events and normalize them into a standard event model.
Microsoft-focused connectors	M365/Entra/Defender initial connector patterns or import stubs are working.
Alert queue	Analyst can view, filter, assign, prioritise and convert alerts into incidents.
Case management	Incident can move from New to Triage to Investigating to Contained to Closed.
AI summarisation	AI can summarise the alert and draft an investigation note with citation to available evidence.
Reporting	Monthly customer report and incident closure report can be generated.
Audit trail	Admin, analyst and playbook actions are logged with user, timestamp and tenant.

Reference standards and sources

NIST Cybersecurity Framework 2.0: <https://csrc.nist.gov/pubs/cswp/29/the-nist-cybersecurity-framework-csf-20/final>

NIST SP 800-61 Revision 3 Incident Response Recommendations: <https://csrc.nist.gov/pubs/sp/800/61/r3/final>

MITRE ATT&CK Enterprise Matrix: <https://attack.mitre.org/matrices/enterprise/>

OCSF Open Cybersecurity Schema Framework: <https://ocsf.io/>

Sigma Detection Format: <https://sigmahq.io/docs/basics/rules.html>

OASIS STIX/TAXII Cyber Threat Intelligence Standards: https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=cti

CIS Critical Security Controls v8.1: <https://www.cisecurity.org/controls/v8-1>

FIRST Traffic Light Protocol 2.0: <https://www.first.org/tlp/>